

CARDIOLINE

200P

User manual

CE

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CARDIOLINE

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CARDIOLINE

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1. GENERAL INFORMATION

This manual is an integral part of the device and should always be available as support material the operator. Strict compliance with the information contained in this manual is an essential prerequisite for a proper and reliable use of the device.

Have the operator read the manual thoroughly as the information related to the different chapters is only described once.

1.1. Other important information

This manual was written with the utmost care. Should you find any details which do not correspond to those contained in this manual, please inform Cardioline SpA who will correct such inconsistencies as soon as possible.

The information contained in this manual is subject to change without notice.

All changes will be in compliance with the regulations governing the manufacturing of medical equipment.

All trademarks mentioned in this document are property of their respective owners. Their protection is guaranteed.

No part of this manual may be reprinted, translated or reproduced without the manufacturer's written authorisation.

The code relating to this manual is listed below.

Language	Code
ENGLISH	36519137_ENG

2. SAFETY INFORMATION

Cardioline SpA will be held responsible for the safety, reliability and functionality of the devices only if:

1. the assembly operations, modifications or repairs are carried out by Cardioline SpA or by its Authorised Service Centre;
2. The device is used in compliance with the instructions provided in the use manual.

Always contact Cardioline SpA should you wish to connect any devices not mentioned in this manual.



Warnings

- This manual provides important information on proper use and safety of the device. Failure to comply with the described operating procedure, improper use of the device, ignoring the specifications and recommendations supplied, may cause severe physical injuries to the operators, patients and bystanders, or may damage the device.
- No modification of this equipment is allowed.
- To avoid the risk of electric shock, this equipment must only be connected to a supply main with protective earth.
- If in doubt regarding the integrity of the external earth conductor, use the device with its inner battery.
- When connecting the printer to other devices, to avoid undermining the safety of the operator and patient, keep in mind the requirements of IEC 60601-1:2005+A1 chapter 16 and measure the leakage current to confirm that there is no risk of electric shock.
- For the correct operation of the device and for the safety of the operators, patients and bystanders, the device and the accessories must be exclusively connected as outlined in this manual.
- Only use the power supply cable provided with the printer, which must not be used for other equipment.
- To guarantee the safety of the operator and of the patient, the equipment connected to the same line as the device must comply with IEC 60950 or IEC 60601-1 standards.
- There is a risk of explosion. Do not use the device in the presence of flammable anaesthetics.
- There is no safety hazard if other equipment, such as pacemakers or other stimulators, is used simultaneously with the device; however, disturbance to the signal may occur.
- The operation may be adversely affected by the presence of strong magnetic fields such as those produced by electro surgery equipment.
- The use of the device is not recommended in the presence of medical diagnostic imaging equipment such as the Magnetic Resonance Imaging (MRI) or Computerised Axial Tomography (CAT) in the same environment.

- Only use the recommended batteries. Using other types of batteries may cause danger of fire or explosion.
- The internal rechargeable battery is hermetically sealed NiMH and requires no maintenance. Should the battery be faulty, contact Cardioline technical assistance service.
- The low battery warning is designed for the recommended batteries only. Using other types of batteries may lead to a lack of indication resulting in device failure. If the battery is low, connect the device to the electrical mains.
- Do not clean the device or the cables by submersing them in liquid, autoclaving, or steam cleaning. This may cause serious damage to equipment or reduce its lifespan. Using non-specific detergents/disinfectants, failure to comply with the recommended procedures or contact with non-specific materials may cause additional risks to operators, patients or bystanders or may damage the device. Do not sterilise the device or the cable with ethylene oxide gas (EO). Refer to Section 7 for instructions on proper cleaning and disinfection.
- Do not leave the cables unattended in the presence of children as they could be accidentally strangled.



Attention

- Do not pull or stretch the power supply cables as this could result in mechanical and/or electrical failures. The cable should be wrapped in a loop and arranged correctly.
- There are no user-serviceable parts inside the device. The device can only be dismantled by qualified service personnel. Any malfunctioning or defective device must be excluded from use and be checked/repared by qualified service personnel before being reused.
- The device does not require any calibration or special instrumentation for correct use and maintenance.
- When it is necessary to dispose of the device, its components and accessories (e.g.: batteries, cables, electrodes) and/or packaging material, comply with local standards for waste disposal.

Notes

- In order to prevent damage to the device during transportation and storage (when still in its original packaging), comply with the following environmental conditions:

Ambient temperature +5°C to +40°C
 Relative humidity 20% to 90%
 Atmospheric pressure 700 hPa to 1060 hPa

- The device is intended for use in hospitals or doctor's offices and should comply with the following environmental requirements:

Ambient temperature +10°C to +40°C

Relative humidity 50% to 90%

Atmospheric pressure 700 hPa to 1060 hPa

- After using the device on battery power, always reconnect the power cable. This will guarantee that the batteries are recharged automatically the next time the device is used.

3. ELECTROMAGNETIC COMPATIBILITY (EMC)

This device requires particular precautions regarding Electromagnetic Compatibility. It must therefore be installed and commissioned in compliance with the information on Electromagnetic Compatibility contained in this manual.

Portable and mobile radio communication equipment can affect operation of the device.

Using accessories, transducers or cables different than those specified in par. **Errore. L'origine riferimento non è stata trovata.** can increase the emissions or decrease the immunity of the appliance.



Warnings

- This device could generate radio interference or disturb operation of the equipment in the vicinity. Therefore it could be necessary to take measures to mitigate these effects, such as re-directing or repositioning the device or shielding the room.
- The use of accessories and cables other than those recommended by Cardioline may cause an increase in emissions or a lowering in the protection of the system.
- The device must not be used near or superimposed to other equipment. If necessary, check that the device works according to its standard operation.
- Use of this equipment adjacent to or stacked with other equipment should be avoided because it could result in improper operation. If such use is necessary, this equipment and the other equipment should be observed to verify that they are operating normally.
- Use of accessories, transducers and cables other than those specified or provided by the manufacturer of this equipment could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.
- Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the device, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.

Notes

Electromagnetic compatibility during the use of the device is required with the surrounding devices.

An electronic device can generate or receive electromagnetic interference. The electromagnetic compatibility test (EMC) has been carried out on the device in compliance with the international EMC directive for medical equipment (IEC 60601-1-2). This IEC standard has been adopted as a European standard (EN 60601-1-2).

Fixed, portable and mobile equipment for RF communication may affect the protection of the medical equipment. See par. 3.4 for the recommended separation distance between the radio equipment and the device.

3.1. Guidance and Manufacturer's declaration - Electromagnetic emissions

The device is intended to operate in the electromagnetic environment specified below. The customer or the user of the device must guarantee that it is used in this environment.

Emission test	Compliance	Electromagnetic environment – guidance
Radiated radio frequency emissions (RF) CISPR 11	Class B Group 1	The device uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment. The device is suitable for use in all establishments other than domestic environments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Conducted radio frequency emissions (RF) CISPR 11	Class B Group 1	
Harmonics emissions IEC 61000-3-2	Class B	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Complies	

3.2. Guidance and Manufacturer's declaration - Electromagnetic immunity

The device is intended for use in the electromagnetic environment specified below. The customer or the user of the device assure that it is used in such an environment.

Immunity test	Conformity	Compliance level	Electromagnetic Environmental Information
Electrostatic discharge (ESD) IEC 61000-4-2	in contact +/- 8 kV airborne +/- 15 kV air	in contact +/- 8 kV airborne +/- 15 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30 %.
Electrical Fast Transient/ Burst IEC 61000-4-4	+/- 2 kV for power lines +/- 1 kV for input/output lines	+/- 2 kV for power lines +/- 1 kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	+/- 0.5, 1 kV line-to-line +/- 0.5, 1, 2 kV between phase and earth	+/- +/- 0.5, 1 kV line-to-line +/- +/- 0.5, 1, 2 kV between phase and earth	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruption and voltage variations on power supply input lines IEC 61000-4-11	<5% U_T (>95% dip in U_T) for 0.5 cycle 40% U_T (60% dip in U_T) for 5 cycles 70% U_T (60% dip in U_T) for 25 cycles <5% U_T	<5% U_T (>95% dip in U_T) for 0.5 cycle 40% U_T (60% dip in U_T) for 5 cycles 70% U_T (60% dip in U_T) for 25 cycles <5% U_T	Mains power quality should be that of a typical commercial or hospital environment. If the user of the device requires continued operation during power mains interruptions, it is recommended that device be powered from an uninterruptible power supply or a battery.

	(>95% dip in U_T) for 5 s	(>95% dip in U_T) for 5 s	
Power frequency and magnetic field (50/60 Hz)	30 A/m at a frequency of 50 and 60 Hz	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

NOTE: U_T is the AC mains voltage prior to the application of the test level.

3.3. Guidance and Manufacturer's declaration - Electromagnetic immunity

The device is intended for use in the electromagnetic environment specified below. The customer or the user of the device should assure that it is used in such an environment.

Emission test	IEC 60601 test level	Compliance level	Electromagnetic Environmental Information
Conducted RF IEC 61000-4-6	3 V rms From 150 kHz to 80 MHz With amplitude modulation 80% 1 kHz sinusoidal waveform	3 V	Portable and mobile RF communications equipment should be used no closer to any part of the ECG100L/ECG200L, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance: $d = \left[\frac{3.5}{3V_{rms}} \right] \sqrt{P}$ $d = \left[\frac{3.5}{3V/m} \right] \sqrt{P} \quad \text{From 80 MHz to 800 MHz}$ $d = \left[\frac{7}{3V/m} \right] \sqrt{P} \quad \text{From 800 MHz to 2.5 GHz}$ where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey (a), should be less than the compliance level in each frequency range (b). Interference may occur in the vicinity of equipment marked with the following symbol: 
Radiated R IEC 61000-4-3 (for RF wireless communications equipment 385 ÷ 5785 MHz)	See table in the standard	Complies with all test levels	
Radiated R IEC 61000-4-3	3 V/m From 80 MHz to 2.7 GHz Modulation amplitude sinusoidal waveform 80% 1 kHz (Professional facilities) 10 V/m From 80 MHz to 2.7 GHz (Homecare Facilities)	3 V/m	

NOTE 1: At 80 MHz and 800 MHz, the higher frequency range applies.

NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

a) Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the device is used exceeds the applicable RF compliance level above, the device should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the device.

b) Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

3.4. Recommended separation distances between portable and mobile RF communications equipment and the device

The device is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the device can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the device as recommended below, according to the maximum output power of the communications equipment.

Maximum rated power output of the transmitter (W)	Separation distance according to frequency of transmitter (m)		
	150 KHz to 800 MHz	800 MHz to 2.5 GHz	800 MHz to 2.5 GHz
	$d = [\frac{3.5}{V_1}] \sqrt{P}$	$d = [\frac{3.5}{E_1}] \sqrt{P}$	$d = [\frac{7}{E_1}] \sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.67	11.67	23.33

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

NOTE 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

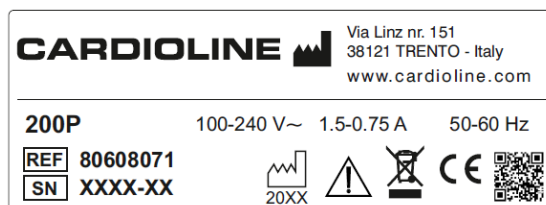
NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

4. SYMBOLS AND LABEL

4.1. Explanation of the symbols

Symbol	Description
	Comply with the instructions in the use manual
	CE marking – compliance with the European Union directives
	Manufacturer
	Reference number (product code)
	Serial Number
	Year of manufacture
	Separate collection of electrical waste and electronic equipment
	Temperature variation
	Humidity variation
	No latex
	Keep dry

4.2. Device label



5. INTRODUCTION

5.1. Purpose of the manual

This manual deals with the 200P device.

The manual represents a guide for the execution of the following operations:

- Reasonable use of the device.
- Preparation of the device for use. (Section 6)
- Troubleshooting and maintenance. (Section 7)

5.2. Description of the device

The device is a thermal printer that is able to print out 12-lead ECG traces on A4 graph and Letter paper.

The device features a type C USB port for connection to a Windows computer.

200P can be powered with an internal battery or from the power mains.

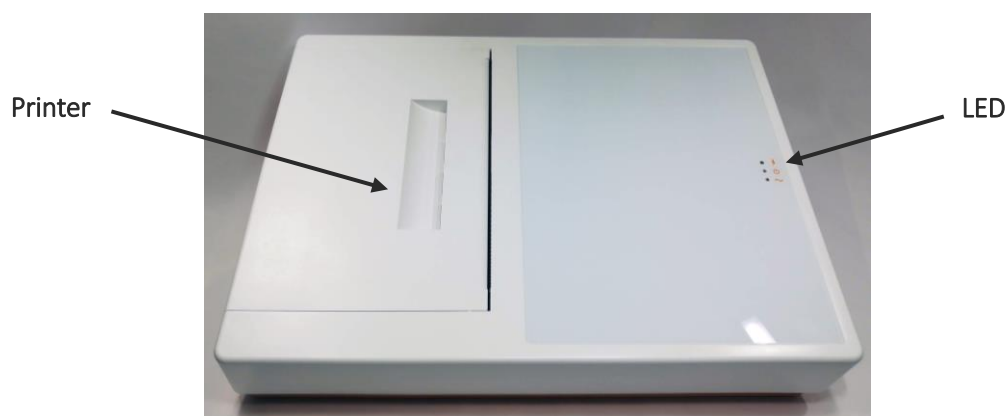
The printer was designed to work with the Cardioline Cubestress and TouchECG applications. It also comes with a printing driver so that it can be used from any Windows application able to operate with Windows-compatible printers.

The device includes:

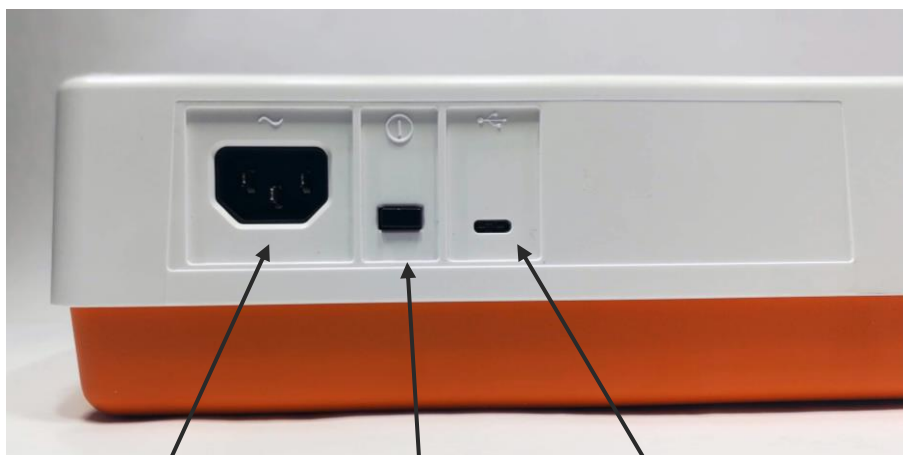
1. Power supply cable
2. Paper
3. User manual

5.2.1. General overview

Front view:



Rear view:



Power supply

On/off
switch

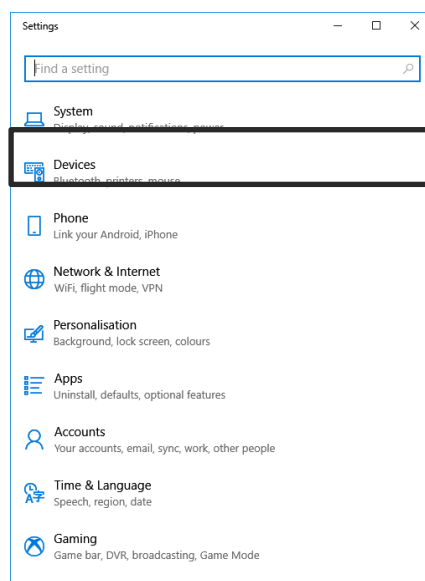
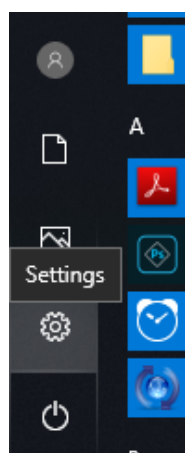
USB

6. PREPARATION FOR USE

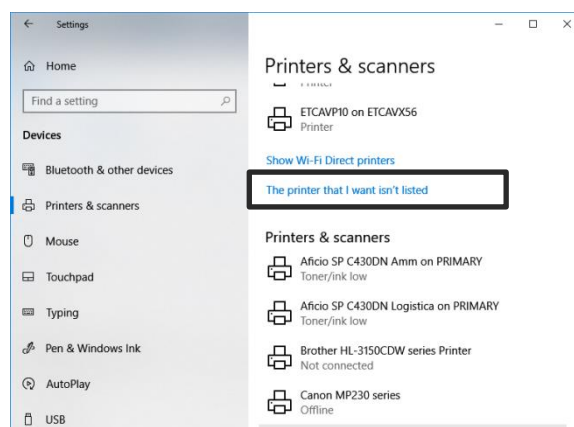
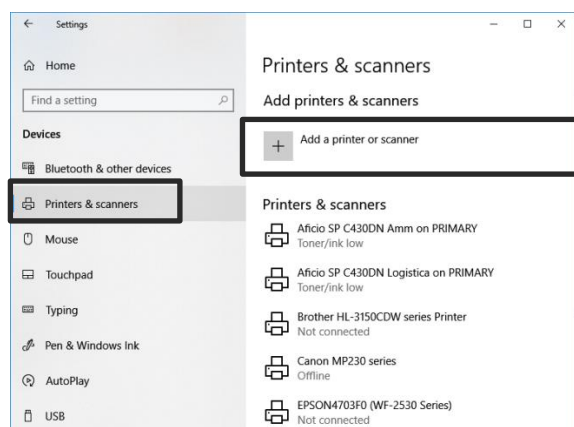
6.1. Installing the printer

Before connecting the printer to the PC, install the printing driver.

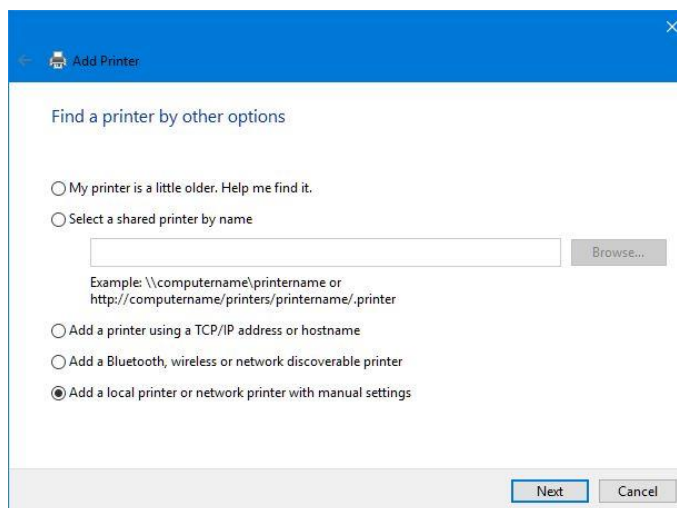
1. From the Windows Start menu, select “Settings” and then “Devices”;



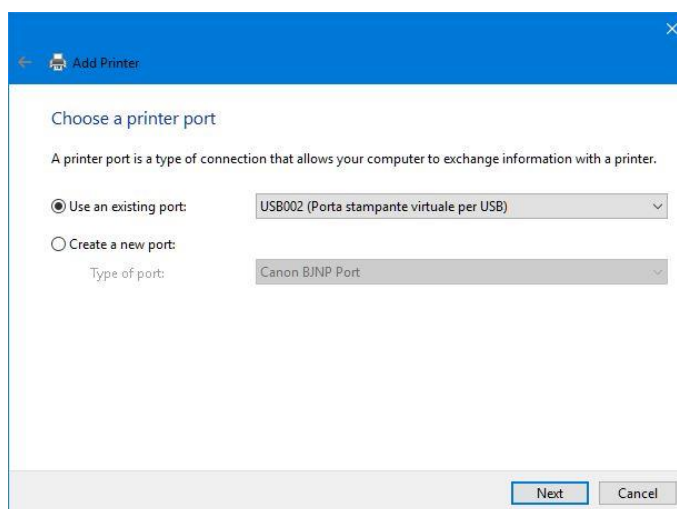
2. Select “Printers and scanners” and then “Add a printer or a scanner”; the computer will start searching for the printer and at the end of the search select “The printer that I want isn’t listed”



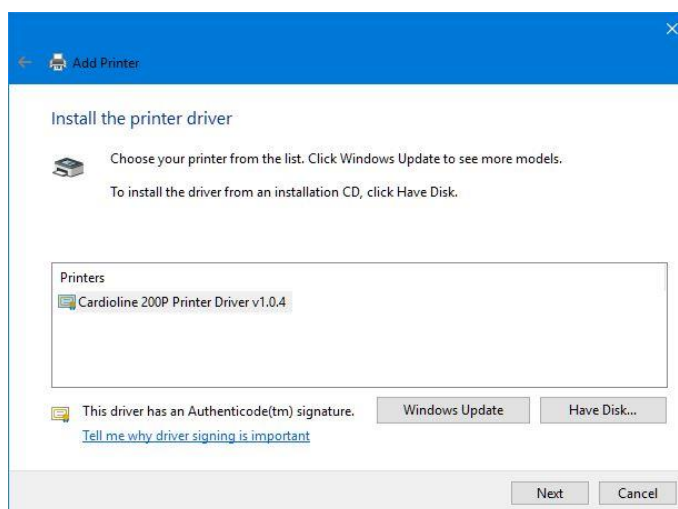
3. Select “Add local or network printer with manual settings”;



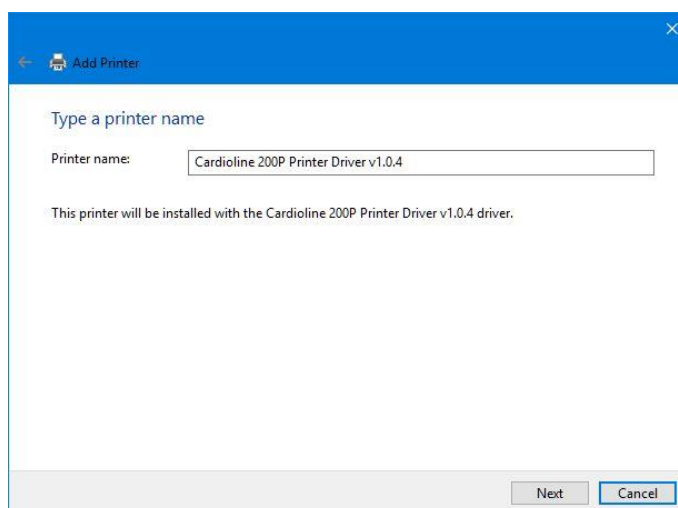
4. Select “Use an existing port” and choose “USB001” from the drop-down menu;



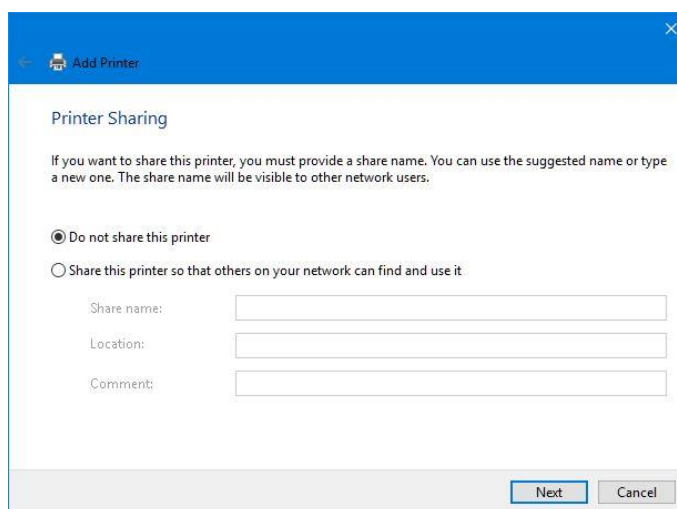
5. Insert the supplied CD or USB drive, select “Driver Disc”, select the “Cardioline” driver and click “Next”;



6. Accept the suggested printer name and click “Next”;



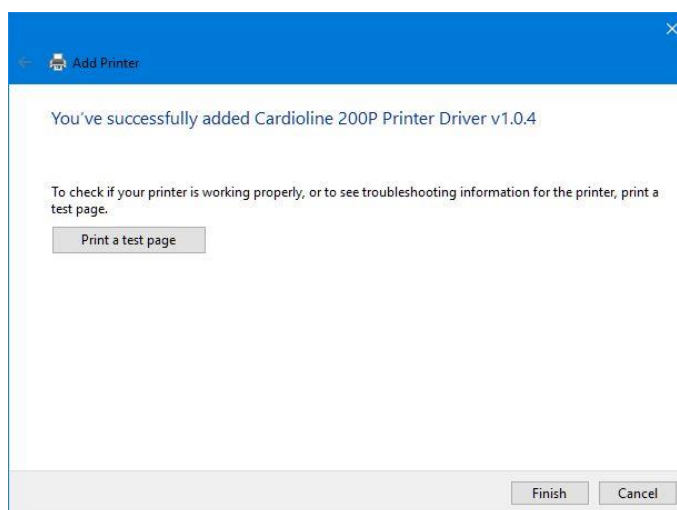
7. Select “Always trust software ...” and click “Install”;
8. Select “Do not share this printer” and click “Next”;



9. Connect the printer to the computer with a USB cable and turn the printer on.

10. After loading in the paper, press “Print test page”

11. Click “End”.

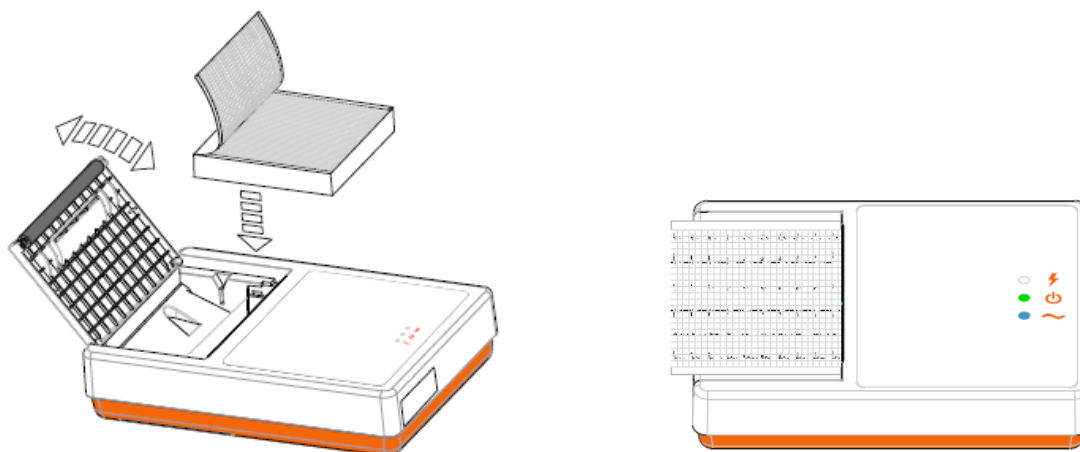


6.2. Loading the paper

Load paper in the device as follows:

1. Remove the outer packaging of the ream of paper.
2. Open the printer compartment, lifting the panel and turning it anticlockwise around its pivot.

3. Slip the roll of thermal paper into the drawer of the printer so that the side of the paper grid faces upwards.
4. Lift the first edge of paper, flip it over (so that the unprinted part faces upwards) and push it to the right so that the edge rests on the right-hand side of the drawer.
5. Close the panel so that the edge of the sheet comes out on the right. The panel clicks when closed properly.



Sequence for loading paper

WARNING: Risk of injuring fingers while handling the paper panel of the printer or the roll control mechanisms.

NOTE: after printing it is important that the paper is cut by pulling it to the right instead of to the left. If the paper is pulled towards the left the door may accidentally open, thus causing problems with any subsequent prints.

NOTE: For the best performance of the thermal printer, use the thermal paper recommended by Cardioline.

6.3. Switching the device ON/OFF

Use the switch at the back of the device to switch the device on and off.

6.4. Power connection

200P can also be powered by battery, disconnected from the mains.

Make sure that the power supply is connected to an earthed electrical socket.

If powered by electrical mains the plug of the main supply is the main switch used to disconnect the device from the main supply. Please, be sure to keep it near the device to simply disconnect the device from main supply.

When the device is not working, it should be plugged in to charge the battery.

NOTE: The first time it is used, the battery must still be fully charged, connecting the device to the power supply.

On top of the device there are three LED lights which show whether and how the device is powered:



Led Light	Description
	Blue – ON, if the device is connected to the mains power supply OFF – when the device is not connected to the mains power supply.
	Green – ON when the device is on. OFF – when the device is off
	Blue – Lights on while device is connected to the electrical mains and battery is charging. Lights off when battery is completely charged or the device is not connected to the electrical mains.

NOTE: There are specific settings which make it possible to extend the life of the battery. Appropriate use and maintenance also extend its life.

6.5. Battery operation

When the device is not plugged in, it is powered by battery.

The device switches off automatically when the battery reaches the minimum allowed level.

NOTE: The first time it is used, the battery must still be fully charged, connecting the device to the power supply.

NOTE: When the device is not working, it should be plugged in to charge the battery.

NOTE: if you use the device in battery mode, always remember to recharge the battery completely after use in order to be sure that the unit is ready for use.

7. MAINTENANCE AND TROUBLESHOOTING

7.1. Precautions

- Switch the device off before inspecting or cleaning it.
- Do not immerse the device in water.
- Do not use organic solvents, ammonia-based solutions or abrasive cleaning agents that could damage the surface of the device.

7.2. Switching off the device

To switch the device off completely, disconnect the power cable and keep the ON/OFF key pressed. Always perform this operation before proceeding with authorised repairs of the device.

7.3. Regular maintenance

Cardioline suggests to regularly check the device:

- Perform a functional and operation check daily
- Periodically clean the unit
- Periodically clean the printer and the printer head
- For use in medical facilities, periodically check the electrical safety of the device
 - Leakage current in the chassis
 - Leakage current to earth
 - Dielectric strength (supply line)

NOTE: Frequency depends on local regulatory requirements and use of the device.

7.3.1. Functional check

Check the device daily before putting it into function:

- Inspect the container and frame for any damage.
- Check the cables and connections for any visible damage.
- Make sure the power button is working correctly and is adequately legible.
- Check battery works properly.

If you notice anything needing a repair, contact an authorised assistance operator to perform it.

7.3.2. Device cleaning

Unplug the device. Clean the external surface of the device with a damp lint-free cloth, using a neutral detergent diluted with water. Thoroughly dry with a clean cloth or paper napkin after washing.

The cleaning solutions allowed are:

- 90% Ethyl alcohol solution
- 90% Methyl alcohol solution
- 90% Isopropyl alcohol solution - only for plastic parts and not for the display/keyboard area.
- Hydrogen Peroxide Solution at 10V
- Hydrogen Peroxide Solution at 36V
- 2% sodium hypochlorite solution

ATTENTION: *Improper cleaning products and operations may damage the device, render the terminals and cables fragile, corrode the metal and invalidate the warranty. Use caution and adopt the correct procedures when cleaning and checking the device.*

7.3.3. Cleaning the thermal head of the printer

- Open the printer cover.
- Rub the printer head delicately with a cloth soaked in alcohol.
- Pass a clean cloth to remove alcohol residue.
- Let the head air-dry.
- Clean the plate using adhesive tape. Apply the tape and lift it up. Turn the roll and repeat the operation until the entire roll is clean.
- Clean the sensor photocell.

7.3.4. Operation check

After cleaning and checking the device, it is possible to check correct operation of the unit by using compatible software to print a standard 12-lead ECG of known amplitude. Printing must be clear and uniform on the whole page. The printing head must not show signs of malfunctioning (e.g. interruptions during printing in the form of horizontal stripes). The paper must slide well and uniformly during printing. The tracings must appear normal, with appropriate amplitude and without distortion or excessive disturbances. The paper must stop with the perforations near the tear bar (indicating that the reference sensor works properly).

7.4. Recommendations

Ensuing any type of assistance on the device or should you suspect non-conforming operation, we recommend the following procedures:

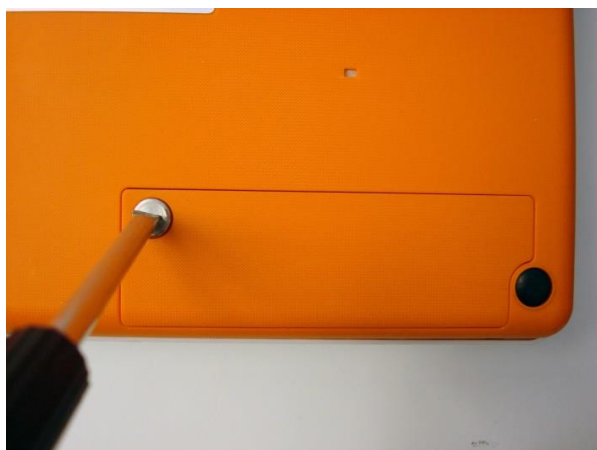
- Check that it works properly.
- For use in medical facilities, periodically check the electrical safety of the device
 - Leakage current in the chassis
 - Leakage current to earth
 - Dielectric strength (supply line)

7.5. Battery maintenance

From the moment the device is installed, the battery lasts approximately 6 months without being recharged. If a low battery is put away for a long period of time, it might not be possible to recharge it.

The battery can be replaced by means of a screwdriver. The battery door is on the bottom side of the device, so to replace the battery follow this procedure:

1. Turn the unit upside down to access the battery door
2. Open the battery door using a screw driver
3. Disconnect the battery connector from the device and take the battery out of its compartment
4. Take a new battery and connect it to the battery connector
5. Place the battery in the battery compartment. Make sure the battery is in place.
6. Close the battery door with the screwdriver.
7. Turn on the device to check if the battery has been replaced correctly. If the device does not turn on, the battery may have been connected in the wrong way. Open the battery door again and repeat the above procedure.



1. Open the battery door with a screw driver.



2. Disconnect the battery connector.



3. Connect the new battery.



4. Place the battery in its compartment.

It is recommended to plug the device in when possible to charge the battery to a maximum.

In any case, the user should try to recharge the battery before it goes flat.

The life of the battery varies based on operating procedures. For better functionality, it is good practice to keep the device plugged in when not used to fully recharge the battery after each use.

When the battery charger reaches the minimum level (10.6V), the device switches off automatically. It takes 4 hours to recharge the battery from its lowest level to 85%. 7 hours are necessary to reach 90%. More time is needed to reach 100%.

The device can be plugged in and used normally even when the battery is charging.

NOTE: every 6 months a full charge/discharge cycle should be performed in order to guarantee a long battery life.

7.6. Troubleshooting table

Problem	Cause	Solution
Electrical interference	Devices generating electromagnetic noise nearby (X-ray devices, magnetic resonance, etc)	Verify there are no devices that can generate interferences nearby.
Device shutting down unexpectedly	Trying to use or switch on the device in battery mode but the battery is depleted	If the device is powered in battery mode, plug it in the supply mains and try to switch it on again.
Noisy printer	Door not completely closed	Open the door and close it firmly.
Paper sliding problems	- Non original paper - Paper placed incorrectly	Verify Cardioline paper is in place. Take out paper and position it as described in this manual.
Device not printing	Incorrect installation/invalid printer port	Go to printer properties, display the "Port" TAB and select the next available USB00x port

7. MAINTENANCE AND TROUBLESHOOTING

		then click "Apply". Any printing queue is immediately serviced, otherwise select a new USB00x port.
Device not printing	▪ Paper is finished	Verify paper is in place. If the exam is not automatically printed at the end of the acquisition, try to print it manually. If manual print does not work contact Cardioline customer service. Verify auto-print settings.
Block of the device	▪ Software problem	Press the On/Off button for at least 10 seconds to restart the device.
The touchscreen doesn't work	▪ Probable loss of touchscreen calibration	Disconnect the battery and the mains for at least 2 hours, then connect the battery and turn on the device. The touchscreen calibration procedure will appear on the display.

8. TECHNICAL SPECIFICATIONS

Thermal printer	8 dot/mm - 216mm; A4/Letter z-fold
Automatic print speed	5, 10, 25, 50 mm/s
Power supply	Medical grade AC power supply (100-240 1.5-0.75 A VAC 50/60 Hz); internal rechargeable battery (NiMH)
Internal battery	Recharging time: 4 hours to 85% of full charge Duration: over 2 hours in continuous printing or more than 500 ECGs
Dimensions	413x295x80mm
Weight	4170 g (with roll of paper)

8.1. Harmonised standards applied

STANDARD	DESCRIPTION
EN 60601-1	Medical electrical equipment - Part 1: General requirements for basic safety and essential performance
EN 60601-1-2	Medical electrical equipment - Part 1: General requirements for basic safety and essential performance - Collateral standard: Electromagnetic compatibility - Requirements and tests

9. WARRANTY

Cardioline SpA guarantees this equipment to be free of defects in material and workmanship for 24 months from date of purchase of the device and for 3 months for spare parts and accessories. The date of purchase shall be proven by a document, issued upon delivery, which shall be submitted in the case of any claim under the warranty.

The warranty provides for free-of-charge repairing or replacement of the equipment parts with manufacturing or material defects. The possible replacement of the equipment is at the manufacturer's discretion. Extended warranty after repairing is not available.

This warranty does not cover defects resulting from:

- tampering, third party negligence, including servicing or maintenance by unauthorised personnel;
- failure to comply with the usage instructions, improper use or use of the equipment different than that for which it was intended;
- improper operation of the power supplies;
- damages caused by fires, explosions or natural disasters;
- use of non-original consumable parts;
- transportation carried out without any precautionary measures;
- use of software programs not associated with the primary function of the machine;
- other circumstances not attributable to manufacturing defects.

Unless otherwise specified, the removable parts, the accessories and the parts which are subject to normal wear are excluded under the warranty; for example: batteries, connection cables, glass parts, computer supports, ink cartridges, etc.

Cardioline Spa declines all liability for any damage which may be caused, directly or indirectly, to persons or property as a consequence of non-compliance with all the prescriptions specified in the manual, especially warnings regarding installation, safety, use and maintenance of the equipment, as well as non-operation of the equipment.

In the event of repair and/or replacement of the equipment or its spare parts, take the equipment to the nearest Cardioline Spa authorised service centre or send it to Cardioline S.p.A. All costs of material and labour will be free of charge and transport costs shall be at the customer's expense.

After 60 months from the date of purchase of the equipment and 3 months from the date of purchase of the accessories and spare parts, the warranty becomes void and service will be provided charging for the parts replaced and labour costs according to the current rates.

Any derogation from the present warranty conditions shall be valid only if expressly approved by Cardioline SpA

10. DISPOSAL

Pursuant to Italian Legislative Decree no. 49 dated 14 March 2014 "Implementation of Directive 2012/19/EU on Waste Electrical and Electronic Equipment (WEEE)", the crossed-out "wheeled bin" symbol on the medical device indicates that, at the end of its service life, the product must be collected separately from other wastes. Therefore, when disposing of the product at the end of its service life, the user is required to contact the supplier or the manufacturer.

Suitable differentiated collection to allow for the subsequent recycling of the decommissioned device, with environmentally-compatible treatment and disposal, helps to prevent any negative effects on the environment and health and to promote the recycling of the materials from which the device is made.

The illegal disposal of the product by the user entails the application of administrative sanctions envisioned by Italian Legislative Decree no. 22/1997 (Art. 50 and subsequent to the Italian Legislative Decree no. 22/1997).

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